

ECE Probabilistic Systems Analysis

Conditional Probability

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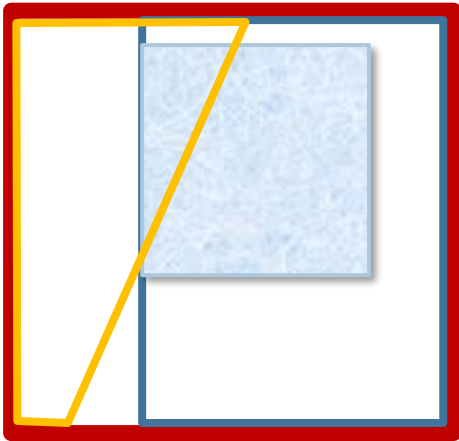


Review of probability models

- Sample space Ω
 - Mutually exclusive
Collectively exhaustive
 - Right granularity
- Event: Subset of the sample space
- Allocation of probabilities to events
 1. $P(A) \geq 0$
 2. $P(\Omega) = 1$
 3. If $A \cap B = \emptyset$,
then $P(A \cup B) = P(A) + P(B)$
 - 3'. If $A_1, A_2, \dots$ are disjoint events, then:
 $P(A_1 \cup A_2 \cup \dots) = P(A_1) + P(A_2) + \dots$

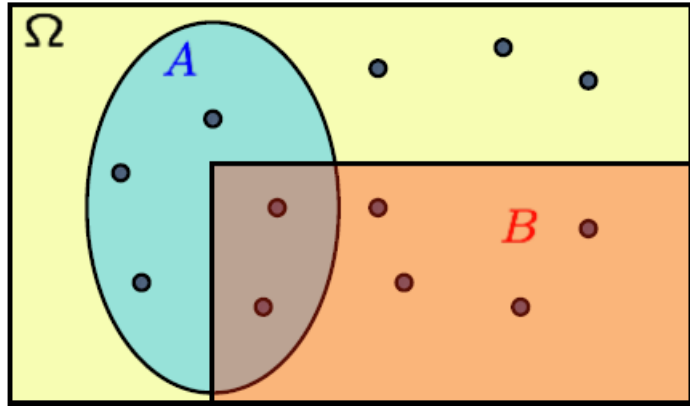
Conditional Probability

Suppose that in a certain city, 30% of the day is rainy, 40 sunny and 60 cloudy. what is the probability that it rains given that it is cloudy



Conditional Probability

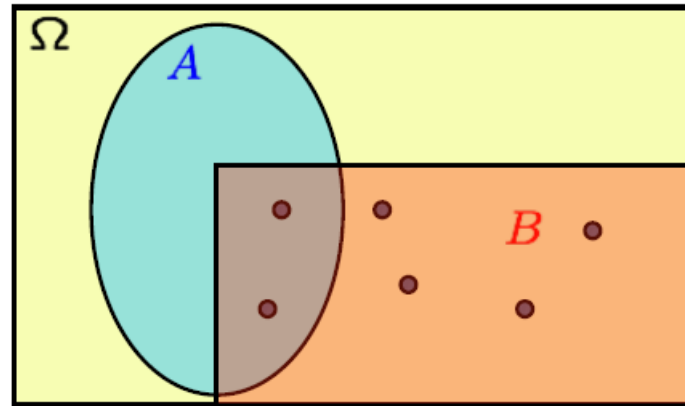
Assume 12 equally likely outcomes



$$P(A) = \frac{5}{12}$$

$$P(B) = \frac{7}{12}$$

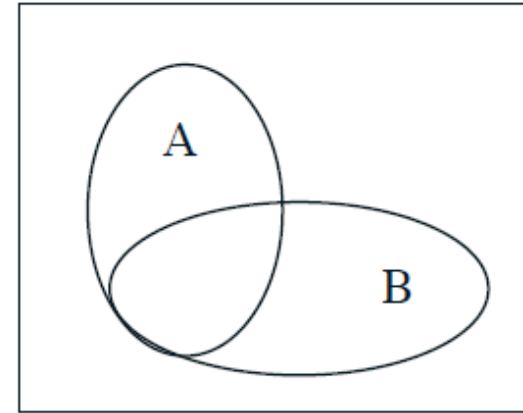
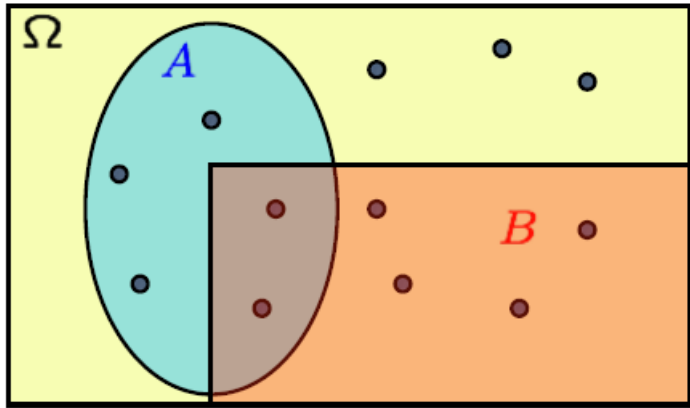
If told B occurred:



$$P(A | B) = \frac{3}{7}$$

$$P(B | B) = 1$$

Conditional probability



- $P(A | B)$ = probability of A ,
given that B occurred
 - B is our new universe
- **Definition:** Assuming $P(B) \neq 0$,

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

$P(A | B)$ undefined if $P(B) = 0$

Example 1.7. A fair 4-sided die is rolled twice and we assume that all sixteen possible outcomes are equally likely. Let X and Y be the result of the 1st and the 2nd roll, respectively. We wish to determine the conditional probability $\mathbf{P}(A | B)$ where

$$A = \{\max(X, Y) = m\}, \quad B = \{\min(X, Y) = 2\},$$

and m takes each of the values 1, 2, 3, 4.

Y = Second roll	4					$P(A B), m=1$
	3					$P(A B), m=2$
	2					$P(A B), m=3$
	1					$P(A B), m=4$
		1	2	3	4	
		X = First roll				

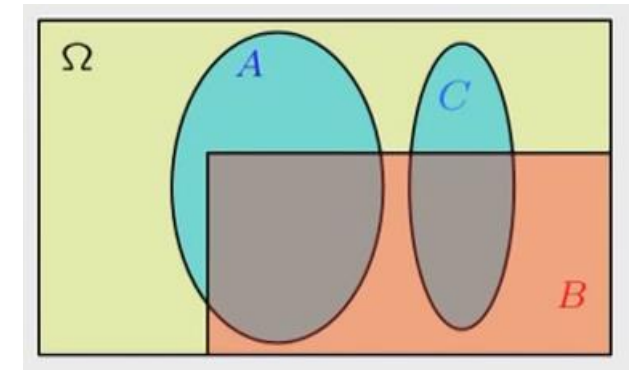
Conditional Probabilities Specify a Probability Law

$$\mathbf{P}(A \mid B) \geq 0 \quad \text{assuming } \mathbf{P}(B) > 0$$

$$\mathbf{P}(\Omega \mid B) = 1$$

$$\mathbf{P}(B \mid B) = 1$$

$$\text{If } A \cap C = \emptyset, \quad \text{then } \mathbf{P}(A \cup C \mid B) = \mathbf{P}(A \mid B) + \mathbf{P}(C \mid B)$$

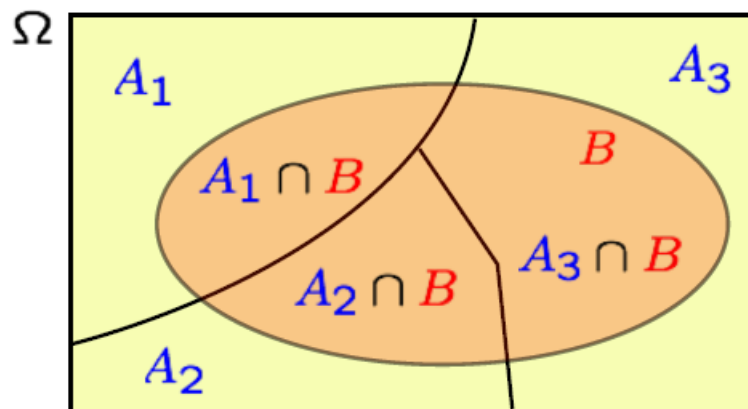


Example Radar detection. If an aircraft is present in a certain area, a radar correctly registers its presence with probability 0.99. If it is not present, the radar falsely registers an aircraft presence with probability 0.10. We assume that an aircraft is present with probability 0.05. What is the probability of false alarm (a false indication of aircraft presence), and the probability of missed detection (nothing registers, even though an aircraft is present)?

Multiplication rule

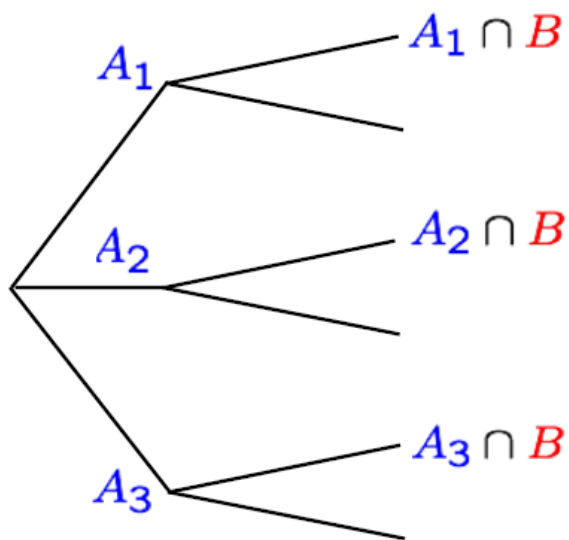
Total probability theorem

$$P(B) = \sum_i P(A_i) P(B | A_i)$$



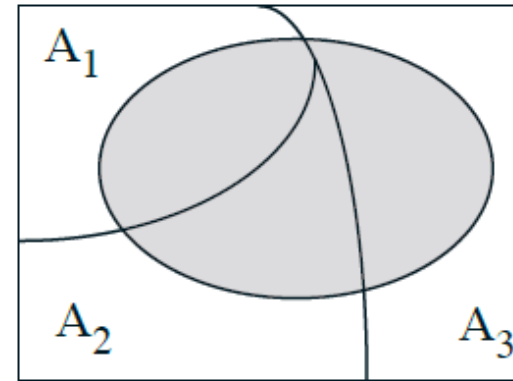
- Partition of sample space into A_1, A_2, A_3
- Have $P(A_i)$, for every i
- Have $P(B | A_i)$, for every i

$$P(B) =$$



Bayes rule

- “Prior” probabilities $P(A_i)$
 - initial “beliefs”
- We know $P(B | A_i)$ for each i
- Wish to compute $P(A_i | B)$
 - revise “beliefs”, given that B occurred



$$\begin{aligned} P(A_i | B) &= \frac{P(A_i \cap B)}{P(B)} \\ &= \frac{P(A_i)P(B | A_i)}{P(B)} \\ &= \frac{P(A_i)P(B | A_i)}{\sum_j P(A_j)P(B | A_j)} \end{aligned}$$

Example:

An urn contains two black balls and three white balls. Two balls are selected at random from the urn without replacement and the sequence of colors is noted. Find the probability that both balls are black and the probability of the event that the second ball is white.

Many communication systems can be modeled in the following way. First, the user inputs a 0 or a 1 into the system, and a corresponding signal is transmitted. Second, the receiver makes a decision about what was the input to the system, based on the signal it received. Suppose that the user sends 0s with probability $1 - p$ and 1s with probability p , and suppose that the receiver makes random decision errors with probability ε . For $i = 0, 1$, let A_i be the event “input was i ,” and let B_i be the event “receiver decision was i .” Find the probabilities $P[A_i \cap B_j]$ for $i = 0, 1$ and $j = 0, 1$.

Which input is more probable given that the receiver has output a 1. Assume that, a priori, the input is equally likely to be 0 or 1

Consider a family that has two children. We are interested in the children's genders. Our sample space is $S = \{(G, G), (G, B), (B, G), (B, B)\}$. Also assume that all four possible outcomes are equally likely.

- a. What is the probability that both children are girls given that the first child is a girl?
- b. We ask the father: "Do you have at least one daughter?" He responds "Yes!"
Given this extra information, what is the probability that both children are girls? In other words, what is the probability that both children are girls given that we know at least one of them is a girl?

In a factory there are 100 units of a certain product, 5 of which are defective. We pick three units from the 100 units at random. What is the probability that none of them are defective?
